

STRUCTURAL CHANGE AND SINGLEHOOD: DECLINING MARRIAGE RATES IN THE SERVICE ECONOMY*

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May 2026

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Abstract

Since 1980, the U.S. has experienced a decline in marriage rates that is driven almost entirely by the less educated. In this paper, we investigate the role of structural transformation in explaining this trend. Structural transformation may alter the gains from marriage, which stem from comparative advantage à la Becker, by changing the wages of women relative to men and by providing market substitutes for home production. We show in an accounting exercise that structural transformation shifts labor compensation away from less educated men and towards less educated women. Then we calibrate an equilibrium model of household labor supply, consumption, and formation and use it to decompose the decline in marriage. In the model, workers specialize in different sectors, including home production, along skill and gender lines. Among low-skilled households, structural transformation reduces the gains from having men specialize in market work and women specialize in home production. In our benchmark calibration, structural transformation accounts for 3.9 percentage points out of the 14.0 percentage point decline in marriage for non-college individuals.

JEL Codes: D13, J12, J16, O14.

Keywords: Structural transformation, marriage, home production, service economy, education.

*The views expressed herein are solely those of the authors and do not necessarily reflect the views of the Federal Reserve Bank of San Francisco or the Federal Reserve System.

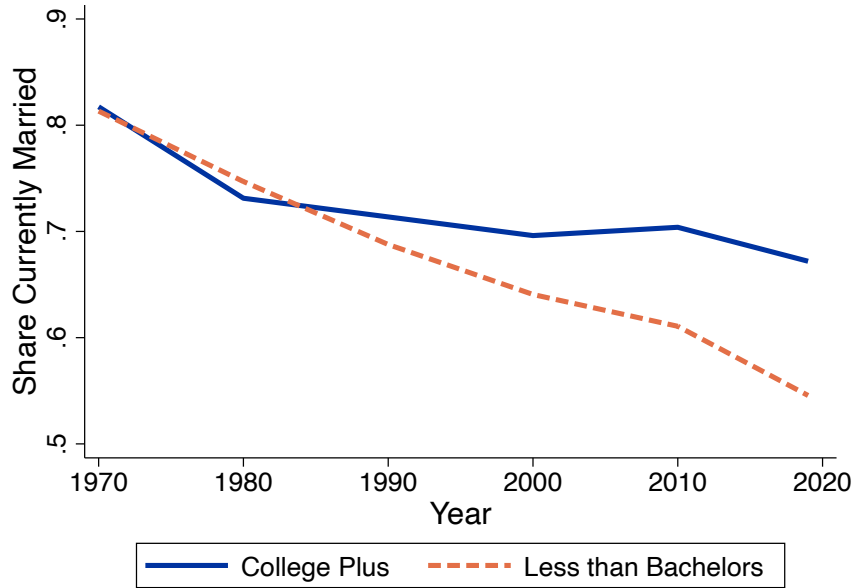
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1 Introduction

In recent decades, Americans have grown more likely to remain single, whether by never marrying or through divorce. This trend has been particularly pronounced among those without a college degree since 1980, leading to a widening education gap in marriage (see Figure 1). As marriage offers benefits such as intra-household risk sharing, economies of scale in public goods, and fertility, both the aggregate decline of marriage and the education gap in marriage are relevant for policy-makers.¹ Over the same period of time, the United States has experienced substantial structural transformation, and in particular, de-industrialization: the long-run shift of employment and production out of manufacturing and into services.

Figure 1: Share of Adults (25-59) Married



Notes: US Census and ACS data from IPUMS (Ruggles et al., 2025). Sample consists of men and women ages 25-59 who report whether they are currently married. Individuals who report at least 4 years of college are considered “College Plus.”

In this paper, we study whether structural transformation plays a role in explaining the decline of marriage in the US, especially its decline among workers without college. We root our analysis in the central insight from Becker (1973): that men and women marry to gain

Acknowledgments: We thank Ingrid Chen for excellent research assistance.

¹See Kearney (2023); Shore (2010); Reeves (2022) for discussions of these economic benefits of marriage.

access to one another’s time in both home and market production. According to Becker, the incentives to marry are stronger when men’s wages are substantially higher than women’s, because the wage gap implies large gains from substituting men’s time in the market for women’s time in home production. Marriage is essential to make this substitution possible, because unlike market production, home production is not tradable across households.

Structural transformation changes the Beckerian marriage calculus in two ways. First, because services use female labor more intensively than manufacturing, the shift of production towards services contributes to a narrowing gender wage gap. Second, some services are close substitutes for home production, and their rising abundance diminishes the importance of home production. Instead of relying on time within the home to provide services like cooking and cleaning, households can more readily trade for these services in the market. Both of these forces —the falling gender wage gap and the abundance of home production substitutes— decrease the gains from having men specialize in market time and women specialize in home production time.

These trends are especially relevant for Americans without a college degree. Market substitutes for home production rely disproportionately on non-college female labor, while production of goods is most intensive in non-college male labor. As households source more services from the market, they raise the opportunity cost of home production for low-skilled women in particular. And as the US economy de-industrializes, it depresses the relative wages of low-skilled men. The upshot is that throughout the rise of market services, the gender wage gap has narrowed fastest for non-college workers, the same group that has seen the steepest decline in marriage.

In the first part of our paper, we document the rapid convergence of labor market outcomes between low-skilled men and women, and attribute a large share of this convergence to structural transformation. Similar to [Ngai and Petrongolo \(2017\)](#), we decompose changes

in the female share of the wage bill into a within-industry and across-industry component.² We do this separately for non-college and college workers, and find that the across-industry component explains the majority of gender convergence among non-college workers, but almost none of the gender convergence among college workers.

We also use data on household expenditures to show that marriage is associated with significant *de-marketization* of home services. Married households spend a smaller share of their resources than single households on home production substitutes like food away from home. This fact supports our conjecture that access to cheaper home production, which households substitute for market output, constitutes part of the gains from marriage. The gap between married and single households' expenditure on home production substitutes has been stable since 1980 for college households, but declined considerably for non-college households. We interpret this decline as evidence that gains from specialization through marriage have shrunk for low-skilled couples.

We then develop a static general equilibrium model of household formation, labor supply, and consumption, and show how structural transformation in the labor market—particularly marketization of home production—affects intra-household allocations as well as the decision to enter into marriage. The model builds on the framework developed in [Ngai and Petrongolo \(2017\)](#), in which the household consumes a bundle of goods and services that includes home production. We incorporate skill heterogeneity into the model, as in [Buera et al. \(2022\)](#). Production in each market sector uses high- and low-skilled labor, and male and female labor, in different intensities. In the calibrated model, low-skilled men specialize in production of goods and low-skilled women specialize in services that substitute for home production, such as food service. High-skilled men and women both specialize in other services that are poor substitutes for home production, such as law and finance.

The key contribution of our model is to endogenize household formation within an otherwise

²[Ngai and Petrongolo \(2017\)](#) do this for the female share of employment, rather than the wage bill.

standard model of structural transformation. Men and women meet in a one-shot marriage market and decide whether to stay single or form a couple. Their decision hinges on the consumption gains from marriage. In line with the Becker model, these consumption gains are driven by *gains from intra-household trade*: marriage increases consumption for both spouses because it allows them to trade home production time and market time with one another. It follows naturally that the size of these consumption gains depends on the importance of home production in the household’s consumption bundle. If home production were irrelevant and all home service needs could be outsourced to the market—i.e. by trading across households—there would be no gains from intra-household trade.

We show that it is possible to use observables to directly measure the consumption gains from marriage. They are a function of relative male and female wages, the share of single households’ resources devoted to home services, and the share of home service value that single households source from the market, relative to married households. Consumption gains are greatest when male and female wages are dissimilar, when home services are an important component of the consumption bundle, and when singles rely on market substitutes for home production much more than married households do—that is, when marriage allows households to de-marketize home production. This is because de-marketization through marriage, which couples traditionally achieved by devoting much of the wife’s time to home production, is an observable measure of the specialization that marriage enables. As the gender wage gap narrows and market substitutes for home production become cheaper, married households see smaller gains from de-marketizing home services. In other words, the transition to a service economy with robust demand for female labor diminishes the material incentives to marry.

We do not assume that marriage is exclusively driven by the gains from intra-household trade. Our model also incorporates an amenity value of marriage that acts as a residual, capturing the benefits of marriage that we do not explicitly micro-found. We use our calibrated model to assess how much of the change in marriage rates is attributable to structural transformation forces, and how much is due to the residual amenity value of

marriage, whose evolution over time may reflect cultural change in attitudes towards marriage. We examine these changes with heterogeneity across skill groups.

More specifically, we calibrate our benchmark economy to match employment shares, time use patterns, and marriage rates in 1980. Then we allow for sector-specific productivity and the amenity value of marriage to evolve over time, in order to match the world in 2010. We allow for this amenity value to follow different trajectories within each skill group; that is, we do not rule out that the emerging education gap in marriage rates could simply reflect cultural shifts. In our model, structural transformation forces alone imply a 3.9 percentage point decline in the marriage rate among low-skilled individuals, along with a 0.8 percentage point increase in marriage for high-skilled individuals. We find that 46% of the widening education gap in marriage rates is thus explained by structural transformation.

2 Related Literature

This paper bridges three economic literatures. First is that on the economic incentives which shape marriage, and the recent decline in marriage rates in the U.S. A number of studies provide empirical evidence that the shift of employment away from manufacturing and towards services—and more broadly a rise in women’s relative earnings—may have caused a decline in marriage rates ([Autor et al., 2019](#); [Moro et al., 2017](#); [Shenhav, 2021](#)). Building on this empirical work, we develop a quantitative equilibrium model which we use to decompose the decline in marriage by skill group and quantify the role of structural transformation, captured by differences in sector productivity growth. Related quantitative approaches by [Chiappori et al. \(2009\)](#); [Fernandez et al. \(2005\)](#) have linked marriage decisions to the rise in inequality, with a focus on positive assortative matching. More closely related to this paper is work by [Greenwood et al. \(2016\)](#), who study the role of falling gender wages gaps and the availability of household durables in the rise of married women’s labor force participation and the decline of marriage. While they argue that the falling prices of household durables

contributes to the aggregate decline in marriage, the effect is similar across education groups. In contrast to their approach, we develop a model of heterogeneous households’ consumption and labor allocations, with equilibrium in a multi-sector labor market. Our model highlights how male and female wages as well as the price of consumption across different goods and services impact the marital surplus, with differential effects across education groups. As in [Becker \(1973\)](#), the marital surplus in our model stems from gains from trade due to gender differences in the opportunity cost of market work.

Second is the literature on convergence in male and female labor market outcomes, especially the rise of married women’s work. Our paper relates to the literature that falling gender wage gaps and declining home production demands explain the rise in women’s—and particularly married women’s—labor force participation ([Attanasio et al., 2008](#); [Bar and Leukhina, 2011](#); [Heathcote et al., 2017](#); [Albanesi and Olivetti, 2016](#); [Greenwood et al., 2005](#); [Rendall, 2024](#); [Olivetti, 2006](#)). Structural transformation provides a unified mechanism through which both changes occur. A closely related paper is [Cerina et al. \(2021\)](#), who relate the rising college premium for women to job polarization via demand for home production substitutes. While they focus on the behavior of single and married individuals in a setting with exogenous marital status, we make the household formation decision central.

Finally, this paper builds on the structural transformation literature. Recent work has highlighted the importance of heterogeneity across gender and education ([Ngai and Petrongolo, 2017](#); [Kuhn et al., 2024](#); [Rendall, 2024](#); [Buera et al., 2019](#); [Chiplunkar and Kleineberg, 2026](#); [Buera et al., 2022](#); [Porzio et al., 2022](#)). We allow for heterogeneity across both dimensions and document that structural transformation accounts for more of the narrowing of gender gaps among low-skill individuals, rather than high skill. A key mechanism in our model is that structural transformation lowers the price of home production substitutable services, a process called “marketization” ([Bridgman and Valentinyi, 2026](#); [Mazzolari and Ragusa, 2013](#); [Ngai and Petrongolo, 2017](#); [Cerina et al., 2021](#)). As this transformation occurs, the “gains from trade” without the household diminish, rendering less important the economic benefit

of marriage.

3 Motivating Facts

While Figure 1 documents the well-known fact that marriage rates have declined more sharply among individuals without a college degree, this section presents additional evidence linking this divergence to structural transformation.

First, we show that since the 1970s, the gender wage gap has converged more rapidly among workers without a college degree than among college-educated workers. This implies a larger decline in women's comparative advantage in home production relative to market work for the less educated, reducing one potential source of gains from marriage.

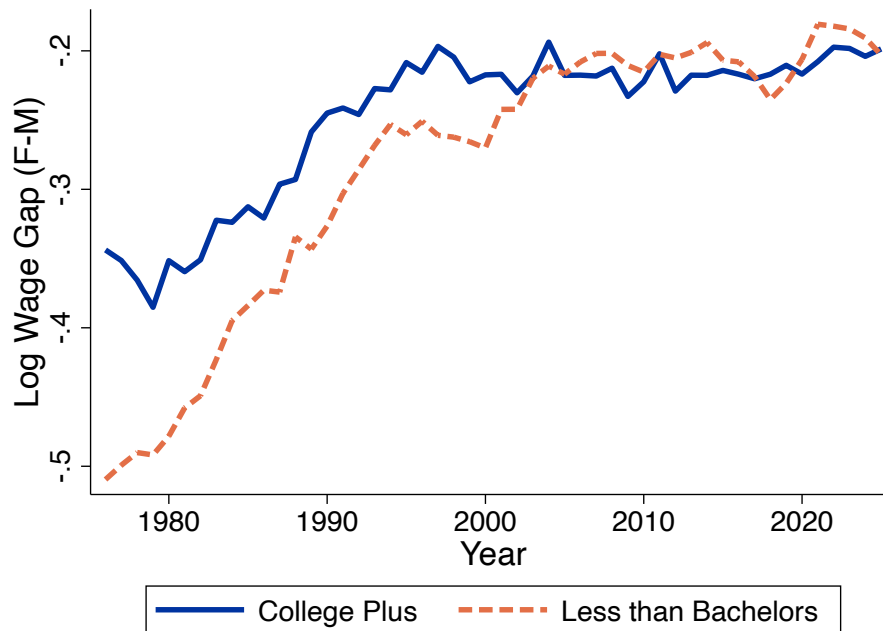
Second, we decompose the rise in women's share of aggregate labor earnings by education group into within-sector and between-sector components. We find that the between-sector component is especially important among workers without a college degree. This suggests that structural transformation is more tightly connected to the convergence in labor-market status between low-skill women and men.

Finally, we show that households differ across education and marital status in the share of consumption expenditure on market substitutes for home production, in line with marriage inducing intra-household specialization in home production. Over time, the consumption expenditure on market substitutes for home production by single and married households converged among the less educated, suggesting that that group was particularly exposed to marketization.

3.1 Rapid Decline of Low-Skill Gender Wage Gap

While the gender wage gap has narrowed considerably since the 1970's, Figure 2 shows this narrowing has been more pronounced among the less educated. The gender wage gap was larger among those without a college degree in the 1970's, but has converged to that of the college educated over time.

Figure 2: Gender Wage Gap by Education Level



Notes: Data from CPS ASEC (Flood et al., 2025). Sample includes civilian wage and salary workers ages 25–54 who worked at least one week in the prior year. Hourly wages are computed as annual wage income divided by annual hours (weeks worked $\times$ usual weekly hours). The top and bottom 1 percent of hourly wages are trimmed within each year. The log wage gap is the difference in mean log hourly wages between women and men, computed separately by education group.

3.2 Structural Transformation and Low-Skill Gender Convergence

The narrowing of gender differences in labor-market outcomes has not come about in the same way across education groups. To show this, we decompose the change in women's share of the wage bill within education groups into a between-industry component and a

within-industry component, following [Ngai and Petrongolo \(2017\)](#).³ The between-industry component captures the extent to which women’s relative labor-market position improves because labor compensation shifts toward industries in which women already account for a larger share of the education-specific wage bill. The within-industry component captures changes in women’s wage bill shares holding fixed the allocation of activity across industries.

Table 1 shows a sharp contrast across education groups. Among non-college workers, most of the increase in women’s share of the wage bill is accounted for by between-industry reallocation: 57.8 percent of the total rise comes from the between-industry component. Among college workers, by contrast, essentially all of the increase comes from within-industry convergence, with only 1.6 percent accounted for by between-industry forces.

This pattern is consistent with structural transformation playing a larger role in reshaping the relative labor-market position of less-educated women and men. The contraction of male-intensive sectors such as manufacturing, together with the expansion of low-skill service industries in which women are relatively more represented, mechanically raises women’s share of labor income among the non-college. By contrast, among college workers, improvements in women’s relative position appear to reflect forces operating primarily within industries rather than sectoral reallocation.

Table 1: Wage Bill Decomposition: 1975–1979 vs. 2015–2019

Skill group	Level			Share of total (%)	
	Total Change (p.p.)	Between	Within	Between	Within
Low-skill (no college)	0.0861	0.0497	0.0363	57.8	42.2
High-skill (college)	0.1807	0.0029	0.1777	1.6	98.4
All workers (pooled)	0.1276	0.0535	0.0742	41.9	58.1

Notes: The between component captures the contribution of reallocation across industries, while the within component captures changes in female wage bill shares within industries. Skill groups are defined by college completion. Estimates are constructed from CPS ASEC data using pooled 5-year initial and final periods.

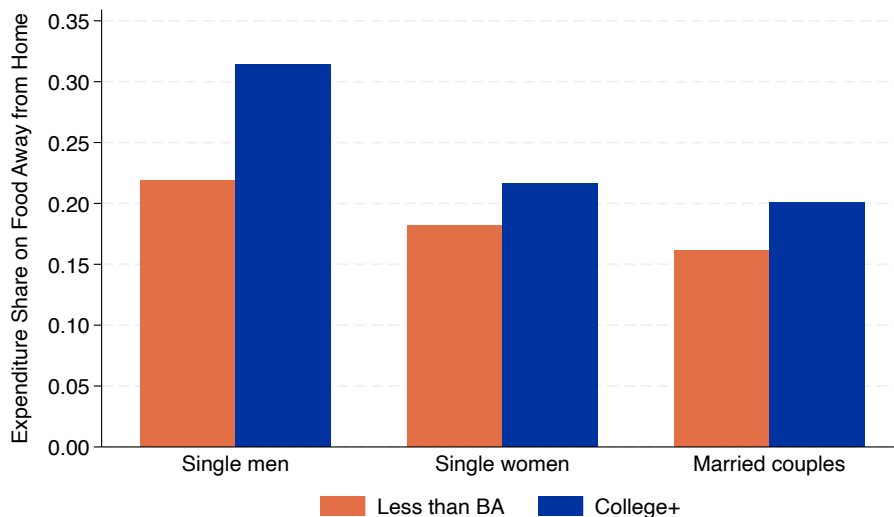
³Industries are grouped into 11 broad categories in the data: agriculture & extractive, construction, manufacturing, transportation & utilities, wholesale trade, retail trade, finance/insurance/real estate, business services, personal services & entertainment, professional services, and public administration. Military is excluded.

3.3 Marriage and Expenditure on Home Production Substitutes

The final motivating fact suggests a link between marketization and marital status. Using the Consumer Expenditure Survey diaries in 1980-1981, we calculate the household expenditure share on food and drinks outside the home, a clear market substitute for home production time. Expenditure shares are reported in Figure 3.

Singles spend more on home production substitutes than married couples do. There is also a clear education gradient: college educated people, for whom the opportunity cost of home production is high, spend more on home production substitutes. These patterns are consistent with two forces which we will model: (1) home production has a higher opportunity cost for workers who face a high wage, and (2) marriage enabling couples to specialize across the market and home sectors, and thus rely less on home production substitutes in the market.

Figure 3: Expenditure on Home Production Substitutes by Household Type

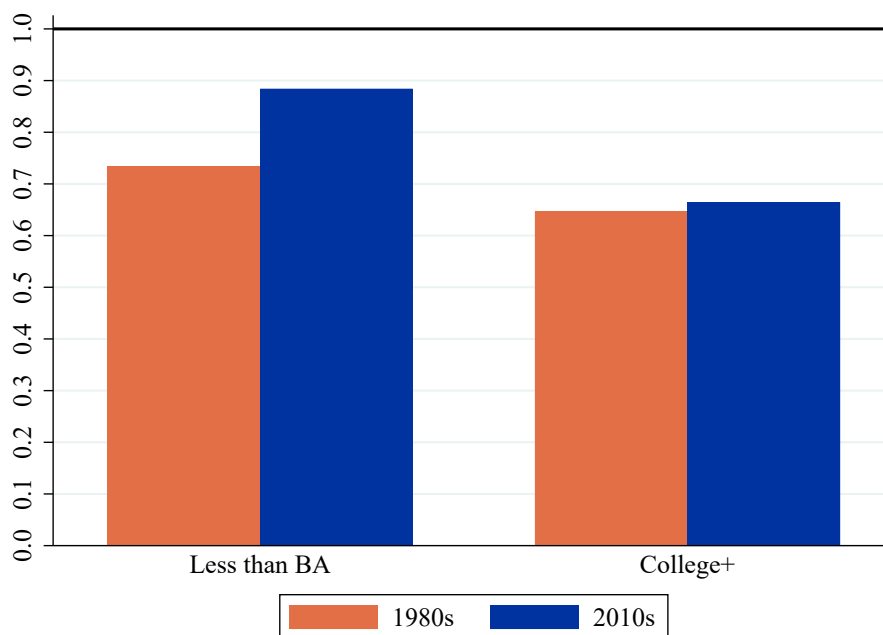


Notes: Data from the 1980-1981 Consumer Expenditure Survey diaries. Sample includes households in which head and spouse are aged 18-65. Expenditure shares indicate the share of total diary expenditure on foods and drinks outside the home. For the married category, education refers to the husband's.

As marketization of home production leads to greater availability and lower prices of market substitutes for home production, this erodes the intra-household specialization benefits of marriage. In Figure 4, we provide suggestive evidence that marketization may have eroded

the gains from marriage among less-educated more than among college-educated. Each bar plots the ratio of the “food away from home” expenditure share for married couples relative to single men. Between 1980 and 2010, the difference in consumption patterns between singlehood and marriage fell among non-college individuals, whereas it remained stable among the more educated.

Figure 4: Married Couple vs. Single Male Relative Expenditure Shares on Home Production Substitutes



Notes: Data from the 1980-1981 and 2010-2014 Consumer Expenditure Survey diaries. Each bar shows the expenditure share on food and drinks away from home for married couples, divided by that of single male households of the same education level. For the married category, education refers to the husband’s.

Together with the evidence in Table 1, these findings suggest that less-educated individuals were especially exposed to the two-pronged erosion of the gains from marriage induced by structural transformation: a narrowing gender wage gap and the expansion of market substitutes for home production.

4 Model

4.1 Environment

The economy is static and populated by a unit measure each of women and men. Individuals differ by gender $g \in \{f, m\}$ and skill level $s \in \{h, \ell\}$, where h_F and h_M denote the aggregate shares of high-skilled women and men, respectively. Individuals form households of size one (single/autarky) or two (married/couple). Households maximize consumption of market goods and services as well as home production.

Each individual has a total time endowment L , which can be allocated between market labor and home production. Individuals earn a wage w_g^s per unit of market labor, which depends only on their gender and skill (thus there are in total 4 different wages in the economy). Home productivity differs across individuals and is denoted by b_g^s , which depends on gender and skill.

Markets are perfectly competitive, meaning both households and firms take wages and prices as given. The economy features markets for final goods and services, labor markets by gender and skill, and a marriage market in which individuals meet potential partners and decide whether to form a household together.

4.2 Production

There are three production sectors: one goods sector (G) and two service sectors. The service sectors are home production substitutes (S) and other final services (O). There is a representative firm in each sector that produces output using labor as its only input, according to a linear technology:

$$Y_j = A_j L_j, \quad j \in \{G, S, O\}, \quad (1)$$

where A_j denotes sector-specific productivity and L_j is an aggregate of labor employed in sector j . Labor used in sector j is in turn a nested CES aggregate of high- and low-skilled labor, which are themselves CES aggregates of female and male labor. Specifically,

$$\begin{aligned} L_j &= \left[\phi_j^{\frac{1}{\lambda}} (L_j^h)^{\frac{\lambda-1}{\lambda}} + (1 - \phi_j)^{\frac{1}{\lambda}} (L_j^\ell)^{\frac{\lambda-1}{\lambda}} \right]^{\frac{\lambda}{\lambda-1}}, \\ L_j^s &= \left[\zeta_j^{\frac{1}{\eta}} (L_{fj}^s)^{\frac{\eta-1}{\eta}} + (1 - \zeta_j)^{\frac{1}{\eta}} (L_{mj}^s)^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \end{aligned} \quad (2)$$

where $s \in \{h, \ell\}$. The parameter ϕ_j governs the intensity with which sector j uses high-skilled labor, while ζ_j governs the intensity with which it uses female labor within a given skill group.

4.2.1 Wages and Prices

Following the properties of the CES production function, the sector-skill wage index is a composite of female and male wages:

$$w_j^s = \left[\zeta_j (w_f^s)^{1-\eta} + (1 - \zeta_j) (w_m^s)^{1-\eta} \right]^{\frac{1}{1-\eta}}, \quad (3)$$

and the overall sectoral wage for composite labor is:

$$w_j = \left[\phi_j (w_j^h)^{1-\lambda} + (1 - \phi_j) (w_j^\ell)^{1-\lambda} \right]^{\frac{1}{1-\lambda}}. \quad (4)$$

Under perfect competition, firms make zero profits and sectoral prices equal unit costs (i.e., wages). Using the linear production technology, the price of output in a given sector is

$$P_j = \frac{w_j}{A_j}, \quad j \in \{G, S, O\}. \quad (5)$$

4.3 Households: Consumption and Time Allocation

Individuals, denoted by n , live in households, denoted by i . They derive utility from consumption of market goods and services, as well as from home services, which can be produced either using market substitutes or time spent in home production. Households differ by size and composition, which affects both the set of feasible allocations and the effective prices they face. We denote household structure by $x_i \in \{c, a\}$, where c stands for a couple (marriage) and a stands for autarky (singlehood).

4.3.1 Preferences

Individual n 's utility is the sum of log consumption and a taste shock ε_i over their household structure x_i .

$$U = \log C_{n,i}^{x_i} + \varepsilon_i(x_i). \quad (6)$$

We place more structure on $\varepsilon_i(x_i)$ below. For now we focus on consumption, which depends in part on household structure.

Household i has total consumption denoted C_i , which in the case of a married couple is split between spouses. For a coupled household with $n \in \{F, M\}$, $C_{F,i}^c + C_{M,i}^c = C_i$. We describe how couples split total consumption below.

Total household consumption C_i is a CES aggregate over consumption of goods (G), home services (HS), and other final services (O):

$$C_i = \left(\beta_G^{\frac{1}{\sigma}} C_{G,i}^{\frac{\sigma-1}{\sigma}} + \beta_{HS}^{\frac{1}{\sigma}} C_{HS,i}^{\frac{\sigma-1}{\sigma}} + \beta_O^{\frac{1}{\sigma}} C_{O,i}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad \beta_G + \beta_{HS} + \beta_O = 1. \quad (7)$$

We assume that the elasticity of substitution between sectors, σ , is less than 1. Home services are themselves a CES aggregate of market-purchased home service substitutes (S)

and home-produced services (HP):

$$C_{HS,i} = \left(\omega^{\frac{1}{\xi}} C_{S,i}^{\frac{\xi-1}{\xi}} + (1-\omega)^{\frac{1}{\xi}} C_{HP,i}^{\frac{\xi-1}{\xi}} \right)^{\frac{\xi}{\xi-1}}. \quad (8)$$

We assume the elasticity of substitution between S and HP , ξ , is larger than 1, meaning they are gross substitutes.

4.3.2 Time Allocation and Home Production

The household's total HP is a CES aggregate of male and female non-market time with elasticity of substitution $\chi > 1$ (imperfect substitutes). That is,

$$C_{HP,i} = \left(\zeta_h^{\frac{1}{\chi}} (b_{F,i} L_{F,i}^{HP})^{\frac{\chi-1}{\chi}} + (1-\zeta_h)^{\frac{1}{\chi}} (b_{M,i} L_{M,i}^{HP})^{\frac{\chi-1}{\chi}} \right)^{\frac{\chi}{\chi-1}} \quad (9)$$

where $b_{F,i}$ and $b_{M,i}$ respectively denote the efficiency of the household's female and male member in home production. Note that if the household consists of a single person, then the labor input of the opposite gender is zero. For example, an autarkic (single) woman's home production is $C_{HP,F}^a = \zeta_h^{\frac{1}{\chi-1}} b_{F,i} L_{F,i}^{HP}$.

4.3.3 Household Endowment and Budget Constraint

All individuals allocate their time L between market labor, which earns a wage w_n , or home production labor. Market labor for n is then $L_n^{mkt} = L_n - L_n^{HP}$.

The household's total endowment E_i is the total market value of its members' time endowments.

$$E_i = w_{M,i} L_{M,i} + w_{F,i} L_{F,i}. \quad (10)$$

If the household consists of a single person, then the labor endowment of the opposite gender is zero.

Total household expenditure on market commodities cannot exceed the household's market earnings. The household's budget constraint can then be expressed as:

$$\begin{aligned} P_G C_{G,i} + P_S C_{S,i} + P_O C_{O,i} &\leq w_{F,i} (L_{F,i} - L_{F,i}^{HP}) + w_{M,i} (L_{M,i} - L_{M,i}^{HP}) \\ \iff P_G C_{G,i} + P_S C_{S,i} + P_O C_{O,i} + w_{F,i} L_{F,i}^{HP} + w_{M,i} L_{M,i}^{HP} &\leq E_i. \end{aligned} \quad (11)$$

The second line makes clear that the household divides its total endowment E_i between expenditure on market commodities (G , S , and O) and home production, which has opportunity cost $w_{F,i} L_{F,i}^{HP} + w_{M,i} L_{M,i}^{HP}$. Going forward, we refer to this opportunity cost as the household's expenditure on home production, $P_{HP,i} C_{HP,i}$. We emphasize that P_G , P_S , and P_O are common across all households, while the home production price $P_{HP,i}$ is household-specific and depends on the household's composition and wages. Households face common prices for G , S , and O because these commodities are traded in a common market, whereas home production is traded only within the household.

The household's problem The household takes its composition, market wages $\{w_g^s\}$, and market prices P_G, P_S, P_O as given. It allocates its members' time between home production and market labor, and allocates market consumption over the three market commodities G , S , and O . It does so with the objective of maximizing total household consumption C_i as defined by Equation (7), subject to the C_{HS} production function defined by Equation (8), the C_{HP} production function defined by Equation (9), and the budget constraint defined by Equation (11).

Consumption allocation within married households Married households must additionally decide how to divide total consumption C_i between their male and female members, subject to $C_{F,i}^c + C_{M,i}^c = C_i$, where the c superscript indicates the individuals' consumption within a couple. We denote by $C_{F,i}^a$ and $C_{M,i}^a$ the individuals' counterfactual consumption under autarky (singlehood), and assume that each household member's share of married

consumption is proportional to their share of autarkic consumption. This assumption implies that men and women experience the same relative consumption gain from marriage. In particular:

$$C_{F,i}^c = \frac{C_{F,i}^a}{C_{F,i}^a + C_{M,i}^a} C_i; \quad C_{M,i}^c = \frac{C_{M,i}^a}{C_{F,i}^a + C_{M,i}^a} C_i \iff \frac{C_{F,i}^c}{C_{F,i}^a} = \frac{C_{M,i}^c}{C_{M,i}^a} = \frac{C_i}{C_{F,i}^a + C_{M,i}^a}. \quad (12)$$

We note that this allocation can be micro-founded as the solution to an intra-household bargaining problem, when bargaining weights are proportional to autarkic consumption.

4.4 The Marriage Market

There is a one-shot marriage market that occurs prior to production and consumption. The marriage market proceeds in two stages. First, each man and each woman draws one potential match from the population of the opposite gender, according to a matching function described below. Second, the members of the potential couple get a common taste shock over marriage and singlehood, and decide whether or not to form a couple. After the marriage market, an endogenous measure of households are single, an endogenous measure are married, and production and consumption proceed as described above. There is no divorce. Other than the marriage taste shock, there is no uncertainty in the model, so individuals make marriage decisions with certainty about their consumption levels conditional on being married or single.

4.4.1 Matching

Recall that h_F and h_M denote the shares of high-skilled women and men. We introduce a matching function $\mathcal{M}(h_F, h_M)$ that produces measures of potential couples across skill groups. $\mathcal{M}(h_F, h_M)$ is a mixture of perfectly random matching, and perfectly assortative matching. The parameter $\rho \in [0, 1]$ governs the degree of assortative matching on skill. When $\rho = 0$, matching is random; when $\rho = 1$, matching is perfectly assortative subject to population

constraints. In particular, we specify $\mathcal{M}$ as follows:

$$\mathcal{M}(h_F, h_M) = \begin{cases} \rho \min\{h_F, h_M\} + (1 - \rho)h_F h_M & \text{H,H,} \\ \rho (1 - \max\{h_F, h_M\}) + (1 - \rho)(1 - h_F)(1 - h_M) & \text{L,L,} \\ \rho \max\{h_F - h_M, 0\} + (1 - \rho)h_F(1 - h_M) & \text{H,L,} \\ \rho \max\{h_M - h_F, 0\} + (1 - \rho)(1 - h_F)h_M & \text{L, H,} \end{cases} \quad (13)$$

where, for example, a type H,L match denotes a match between a high-skilled woman and a low-skilled man. If $h_F = h_M$ and $\rho = 1$, then all high-skilled women match with high-skilled men, and all low-skilled women match with low-skilled men.

4.4.2 The Marriage Decision

The members of each potential couple consider their individual utility under marriage ($x_i = c$) and under autarky ($x_i = a$). As specified above, utility depends both on log consumption and on a taste shock for marriage or autarky. These taste shocks represent the non-economic component of the couple's match quality. This non-economic component includes emotional reasons for marriage (e.g., love) as well as other benefits of marriage that we do not model, such as a compatible beliefs and considerations related to child-bearing.

The taste shock for marriage has a systematic component $\bar{u}_{s_F, s_M}$ that reflects the average amenity value of marriage between two skill groups s_F and s_M . We allow for this amenity value to differ across skill pairings; for example, marriage may have a greater amenity value on average for two high-skilled people than two low-skilled people, in which case $\bar{u}_{H,H} > \bar{u}_{L,L}$. Taste shocks for both marriage and autarky also have idiosyncratic components that we denote ε_i^c and ε_i^a , respectively. Both members of a potential couple draw the same taste shock, so there is no disagreement over whether to form a couple. For a given draw of these taste shocks, utility for a potential couple of skills (s_F, s_M) under each household structure is

as follows:

$$\begin{aligned}
\text{Married woman: } U_{F,i}^c &= \log C_{F,i}^c + \bar{u}_{s_F, s_M} + \varepsilon_i^c \\
\text{Single woman: } U_{F,i}^a &= \log C_{F,i}^a + \varepsilon_i^a \\
\text{Married man: } U_{M,i}^c &= \log C_{M,i}^c + \bar{u}_{s_F, s_M} + \varepsilon_i^c \\
\text{Single man: } U_{M,i}^a &= \log C_{M,i}^a + \varepsilon_i^a.
\end{aligned} \tag{14}$$

Finally, we assume that ε_i^c and ε_i^a are i.i.d. and follow a Type-I (Gumbel) extreme value distribution with scale parameter $\frac{1}{\gamma}$. Under this parametric assumption, the probability π_i^{mar} that a potential couple i chooses to marry is:

$$\begin{aligned}
\pi_i^{mar} &= \frac{\exp [\gamma (\log C_{F,i}^c + \bar{u}_{s_F, s_M, i})]}{\exp [\gamma \log C_{F,i}^a] + \exp [\gamma (\log C_{F,i}^c + \bar{u}_{s_F, s_M, i})]} \\
\iff \pi_i^{mar} &= \frac{\tilde{u}_{s_F, s_M, i} \exp [\gamma (\log C_i)]}{\exp [\gamma \log (C_{F,i}^a + C_{M,i}^a)] + \tilde{u}_{s_F, s_M, i} \exp [\gamma (\log C_i)]},
\end{aligned} \tag{15}$$

where the second line uses the assumption from Equation (12) that the relative consumption gain from marriage is $\frac{C_i}{C_{F,i}^a + C_{M,i}^a}$ for both individuals in the potential couple. For notational convenience, we denote $\exp(\gamma \bar{u}_{s_F, s_M, i})$ by $\tilde{u}_{s_F, s_M, i}$.

4.5 Equilibrium

Equilibrium consists of measures of 8 types of households, consumption and time allocations for each household type, sectoral output and employment of every gender-skill group, prices for each market sector, and wages for each gender-skill group, such that:

1. Firms minimize production costs taking wages as given, implying the sectoral price indices defined by Equations (3)-(4).
2. Households maximize consumption as defined by Equation (7), subject to the home production function (9) and the budget constraint (11), taking prices and wages as

given.

3. The product market for each market sector clears:

$$Y_j = \int_i C_{j,i} di \text{ for } j \in \{G, S, O\}. \quad (16)$$

4. The labor market for each gender-skill group clears:

$$\sum_{j \in \{G, S, O\}} L_{g,j}^s = \int_i L_{g,i}^{s, mkt} di \text{ for } g \in \{f, m\}, s \in \{\ell, h\}. \quad (17)$$

5. The measure of each household type satisfies the matching function in (13) and the marriage probability, conditional on matching, in (15).⁴

4.6 Analysis

We now analyze the properties of our model in equilibrium, particularly as it relates to marriage rates. We first characterize the solution to the household's problem and the effective prices facing the household. Then we show how these prices govern the economic incentives to marry, and develop a sufficient statistic to measure the *gains from intra-household trade* that motivate marriage in the model.

4.6.1 Solving the household's problem

We use standard properties of the CES demand system for total consumption C_i , home services $C_{HS,i}$, and home production $C_{HP,i}$, to derive the following results on the household's allocation. For a given consumption commodity j , we express the household's consumption allocation in terms of expenditure shares $\vartheta_{j,i} = \frac{P_{j,i} C_{j,i}}{E_i}$. Given prices, these expenditure shares

⁴The 8 possible household types are singles of each gender-skill group, and each marriage combination of two skill groups from opposite genders.

are as follows:

$$\begin{aligned}
\vartheta_{j,i} &= \beta_j \left(\frac{P_{j,i}}{P_i} \right)^{1-\sigma} \quad \text{for } j \in \{G, HS, O\} \\
\vartheta_{S,i} &= \omega \left(\frac{P_S}{P_{HS,i}} \right)^{1-\xi} \quad \vartheta_{HS,i} = \mu_i \vartheta_{HS,i} \\
\vartheta_{HP,i} &= \frac{w_{F,i} L_{F,i}^{HP} + w_{M,i} L_{M,i}^{HP}}{E_i} = (1-\omega) \left(\frac{P_{HP,i}}{P_{HS,i}} \right)^{1-\xi} \quad \vartheta_{HS,i} = (1-\mu_i) \vartheta_{HS,i}.
\end{aligned} \tag{18}$$

We use the character μ_i to represent the *marketization* of the household's home service consumption. Note that μ_i is the share of home service value that household i sources from market substitutes S instead of home production.

The household's allocation of labor between home and market production also follows from the CES nature of home production.

$$\begin{aligned}
L_{F,i}^{HP} &= \frac{1}{w_{F,i}} \zeta_h \left(\frac{w_{F,i}/b_{F,i}}{P_{HP,i}} \right)^{1-\chi} \vartheta_{HP,i} E_i, \quad L_{F,i}^{mkt} = L_{F,i} - L_{F,i}^{HP}, \\
L_{M,i}^{HP} &= \frac{1}{w_{M,i}} (1-\zeta_h) \left(\frac{w_{M,i}/b_{M,i}}{P_{HP,i}} \right)^{1-\chi} \vartheta_{HP,i} E_i, \quad L_{M,i}^{mkt} = L_{M,i} - L_{M,i}^{HP}.
\end{aligned} \tag{19}$$

While households face the same prices for market commodities G, S, O , their total price indices differ because they face different effective prices of home production. This is because the price of home production is in fact the opportunity cost of time forgone in the market, which depends on wages. In particular, the price system facing the household satisfies:

$$\begin{aligned}
P_{HP,i} &= \begin{cases} \left(\zeta_h \left(\frac{w_{F,i}}{b_{F,i}} \right)^{1-\chi} + (1-\zeta_h) \left(\frac{w_{M,i}}{b_{M,i}} \right)^{1-\chi} \right)^{\frac{1}{1-\chi}} & \text{for married HHs} \\ \zeta_h^{\frac{1}{1-\chi}} \frac{w_{F,i}}{b_{F,i}} & \text{for single women} \\ (1-\zeta_h)^{\frac{1}{1-\chi}} \frac{w_{M,i}}{b_{M,i}} & \text{for single men} \end{cases} \\
P_{HS,i} &= \left(\omega P_S^{1-\xi} + (1-\omega) P_{HP,i}^{1-\xi} \right)^{\frac{1}{1-\xi}} \\
P_i &= \left(\beta_G P_G^{1-\sigma} + \beta_{HS} P_{HS,i}^{1-\sigma} + \beta_O P_O^{1-\sigma} \right)^{\frac{1}{1-\sigma}}.
\end{aligned} \tag{20}$$

The total consumption of the household can then be written as $C_i = \frac{E_i}{P_i}$, where $E_i = w_{F,i}L_{F,i} + w_{M,i}L_{M,i}$ and P_i is characterized by the system of prices above.

4.6.2 Marriage and the gains from intra-household trade

The solution to the household problem allows us to show how the relative consumption gains from marriage depend on market prices and wages. We compare married consumption $C_i = \frac{w_{F,i}L_{F,i} + w_{M,i}L_{M,i}}{P_i}$ against autarkic consumption $C_{F,i}^a + C_{M,i}^a = \frac{w_{F,i}L_{F,i}}{P_{F,i}^a} + \frac{w_{M,i}L_{M,i}}{P_{M,i}^a}$, where $P_{F,i}^a$ and $P_{M,i}^a$ respectively denote the price indices faced by the woman and the man if they stay single. The log relative consumption gain is the negative log of a weighted average of price reductions from marriage, with the woman and man weighted according to their potential market earnings:

$$\log \left(\frac{C_i}{C_{F,i}^a + C_{M,i}^a} \right) = -\log \left(\frac{w_{F,i}L_{F,i}}{w_{F,i}L_{F,i} + w_{M,i}L_{M,i}} \frac{P_i}{P_{F,i}^a} + \frac{w_{M,i}L_{M,i}}{w_{F,i}L_{F,i} + w_{M,i}L_{M,i}} \frac{P_i}{P_{M,i}^a} \right). \quad (21)$$

Equation (21) makes clear that the benefits of marriage come through a reduction in the effective price faced by women and men for their consumption bundle. Because single and married households face the same prices for market commodities, the price reduction from marriage operates through a decrease in the price of home production, which is tradable within the household but not in the broader market. Marriage yields economic benefits because it allows previously autarkic women and men to trade with each other, exchanging home production for market earnings that can be used to purchase market commodities. For this reason, we refer to the economic gains from marriage as the *gains from intra-household trade*.

A key result of our model is that it is possible to measure the price reduction from marriage—and hence the gains from intra-household trade—directly from the consumption allocations of single and married households. In particular, the CES demand system implies that the price savings from marriage can be expressed in terms of the home service expenditure

share of single households, and the marketization of single and married households' home service consumption:

$$\begin{aligned}\frac{P_i}{P_{F,i}} &= \left(1 + \vartheta_{HS,F,i}^a \left[\left(\frac{\mu_{F,i}^a}{\mu_i} \right)^{\frac{1-\sigma}{1-\xi}} - 1 \right] \right)^{\frac{1}{1-\sigma}} ; \\ \frac{P_i}{P_{M,i}} &= \left(1 + \vartheta_{HS,M,i}^a \left[\left(\frac{\mu_{M,i}^a}{\mu_i} \right)^{\frac{1-\sigma}{1-\xi}} - 1 \right] \right)^{\frac{1}{1-\sigma}} ,\end{aligned}\tag{22}$$

where $\vartheta_{HS,F,i}^a$ is the woman's home service expenditure share when single, $\mu_{F,i}^a$ is her home service marketization when single, and likewise for $\vartheta_{HS,M,i}^a$ and $\mu_{M,i}^a$ for the man when single. Note that when $\sigma < 1$ and $\xi > 1$, the price reduction for gender g is strongest when $\frac{\mu_{g,i}^a}{\mu_i}$ and $\vartheta_{HS,g,i}^a$ are both large. We show how to derive Equation (22) in our Theoretical Appendix.

Equations (21) and (22) show that our model has empirical content. The gains from intra-household trade via marriage are greatest when singles devote a large share of resources to home services ($\vartheta_{HS,i}^a$ is large), and when marriage results in significant *de-marketization* of home production: when the share of home services that singles delegate to the market, μ_i^a , is much larger than the share that married households delegate to the market, μ_i . From Equation (21), we see that these gains also depend on relative potential earnings between men and women: more weight is given to price reductions for the household member with higher potential earnings. The marital consumption gain is especially high if male earnings significantly outweigh female earnings and men face a higher effective price of home production, forcing them to rely more on market substitutes.

Factors affecting marital consumption gains To illustrate the conditions under which marriage is more attractive, Figure 5 shows comparative statics for the marital surplus in partial equilibrium. Each panel manipulates a price faced by the household and displays how the marital consumption gain, or marriage surplus, responds to that price. The left panel shows that the gains from marriage are higher when the price of home production substitutes

S is high: when the household is less able to substitute for home production in the market, the gains from specialization afforded by marriage are beneficial.

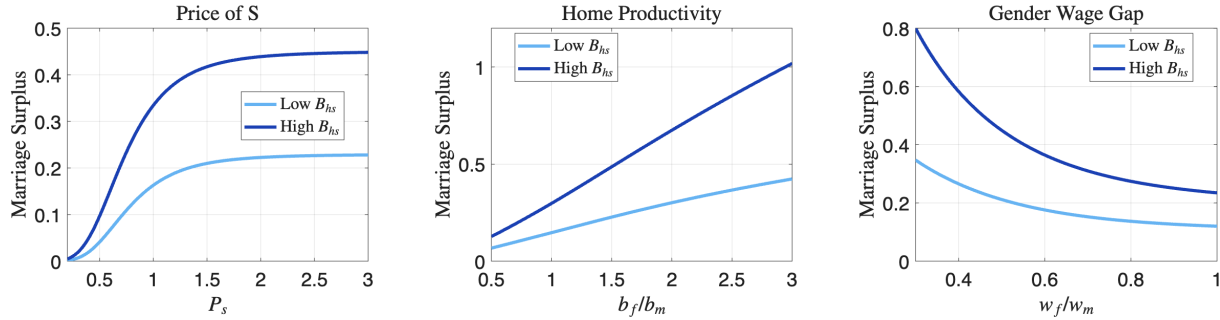


Figure 5: Marital Surplus Comparative Statics

The center panel in Figure 5 varies the relative price of male and female home production by varying the gap in home production productivity b_g . The marriage surplus increases with women's relative home productivity b_f/b_m . Similar to the Heckscher-Ohlin theory of trade, marriage brings greater economic benefits the more it allows women and men to specialize in activities that use their labor most intensively: home production for women, and market production for men.

The right panel in Figure 5 varies the gender wage gap w_f/w_m . A narrower gender wage gap (larger w_f/w_m) lowers the marriage surplus by reducing the gender gap in the opportunity cost of home production. This effect is similar to lower b_f/b_m in the center panel. Increasing w_f/w_m has the additional effect of shifting weight from the male price reduction $\frac{P_i}{P_{M,i}}$ to the female price reduction $\frac{P_i}{P_{F,i}}$ in Equation (21). This effect further reduces the marital surplus in the empirically relevant case where single women rely less than single men on market substitutes for home production. We note that rising productivity in market substitutes – an increase in A_S – can rationalize both a decrease in P_S and an increase in w_f/w_m , especially among low-skilled workers.

In all three panels, the dark line shows the surplus under a high weight on home services β_{HS} in the utility function, whereas the light line represents a low weight. The more resources households devote to home services, the greater the gains from marriage, because

intra-household trade between spouses occurs within that consumption category.

Decomposing the marriage decline With data on consumption expenditures, time use, and wages, it is possible for us to directly measure all components of the gains from intra-household trade $\log\left(\frac{C_i}{C_{F,i}^a + C_{M,i}^a}\right)$, using Equations (21) and (22). These gains are of central importance to marriage rates. Manipulation of the marriage probability in Equation (15) shows that the log odds ratio of marriage for a given potential couple can be decomposed into consumption gains via marriage and the amenity value of marriage:

$$\log\left(\frac{\pi_i^{mar}}{1 - \pi_i^{mar}}\right) = \gamma \log\left(\frac{C_i}{C_{F,i}^a + C_{M,i}^a}\right) + \log \tilde{u}_{s_{F,i}, s_{M,i}}. \quad (23)$$

Our theory therefore allows us to attribute changes in marriage rates to a measurable component—the gains from intra-household trade—and a residual component, $\tilde{u}_{s_{F,i}, s_{M,i}}$.

5 Calibration

Our goal is to decompose the decline in marriage, with special attention to the gap between education groups, into the economic forces driven by structural transformation, which we take to be differential productivity growth across market sectors and an increase in the supply of college-educated workers, as well as non-economic forces (i.e., preferences). We calibrate the model to 1980 as our initial period, and then allow several parameters to evolve: sectoral productivity A_j for $j \in \{G, O, S\}$, high skill shares for men and women (h_F and h_M), and marriage market preferences ($\tilde{u}$'s).⁵ We recalibrate this subset of parameters to 2010 as our later period, keeping all other parameter values fixed at their 1980 levels. Section 5.1 describes the data we use for calibration, while Sections 5.2 and 5.3 describe the calibration steps in detail.

⁵In future iterations of this paper, we intend to endogenize the education decision, as it likely responds to productivity growth across sectors.

5.1 Data

Data for model calibration comes primarily from the 1980 Census and 2010 ACS (Ruggles et al., 2025). We focus on the working-age population aged 22-55 and we define high skill as having at least 4 years of college. We calculate the share of each gender and education group who are single as well as sorting on education within marriage. We also calculate wage bill shares by skill and by gender for each of the sectors in our model. To do so, we impose the following mapping between Census industries and our model sectors: sector G comprises agriculture, mining, manufacturing, construction, transportation, and utilities; sector S includes industries whose output can substitute for home production, which we choose to be food services, personal services, lodging, health care, education, and social services; and O captures the remaining service industries, which are trade, finance, entertainment, professional services, and government. Military and unclassified industries are dropped.

In addition to the Census/ACS, we use the AHTUS from IPUMS to calculate weekly hours spent on market work and on home production, following Ngai and Petrongolo (2017) (Fisher et al., 2018). Since individuals' time endowment is spent on either market work or home production in the model, we abstract from leisure and calculate the share of time on home production by gender, education, and marital status as:

$$\frac{\text{Hours Home Production}}{\text{Hours Home Production} + \text{Hours Market Work}}$$

5.2 Calibration of Model to Early Period

The calibration proceeds in 4 steps. First, we set several parameters exogenously, taking values from related papers in the literature. These are shown in Table 2. We also directly observe the share of men and women with a college degree in the Census, giving us h_M and h_F . The parameter ρ , which governs the extent of positive assortative matching in marriage

market meetings, is difficult to disentangle from the $\tilde{u}$'s, so we set this to 0.6 for now.

Table 2: Exogenous Parameters

Parameter	Value	Source
EOS b/t Skills in Production (λ)	1.41	Katz and Murphy (1992)
EOS b/t Genders in Market Production (η)	2.27	Ngai and Petrongolo (2017)
EOS b/t Genders in Home Production (χ)	2.27	Ngai and Petrongolo (2017)
EOS b/t Sectors in Utility (σ)	0.1	Buera et al. (2022)
EOS b/t S and HP within HS (ξ)	4	Assumption
PAM in Marriage Market Meetings (ρ)	0.6	Assumption

Second, we impose several normalizations. Since ζ_h cannot be separately identified from the b_g^s parameters, we set $\zeta_H = 0.5$. We also set the sectoral productivity terms $A_G = A_O = A_S = 1$ for the early period; we will allow these to grow at different rates when we calibrate to the later period. We normalize $b_m^l = 1$. The male low-skill wage serves as the numeraire for prices. Finally, as in Equation 7, the utility weights across sectors must sum to 1. This leaves the 17 parameters displayed in Table 3 to be calibrated.

In the third step, we jointly calibrate the parameters listed in the top half of Table 3 using method of moments. Notice that these parameters comprise production and utility parameters, which, given the observed composition of households, are independent of the marriage market preferences. We take the observed single and married shares from the data, and calibrate these parameters to match moments pertaining to the sector wage bills by skill and gender, as well as market labor and home production time allocations; the model hits these moments perfectly.

In the final step, we jointly calibrate the marriage market preference parameters listed in the second half of Table 3. Again, we use method of moments, targeting single rates by skill and gender and marriage types. Of special note is the last parameter, the scale parameter of the marriage preference shock (γ). We calibrate this to match the semi-elasticity of marriage to changes in the gender wage gap documented by [Shenhav \(2021\)](#). She finds that over the period 1980-2010, a 10% increase in women's relative wages leads to a 3 p.p. increase in the

Table 3: Calibrated Parameters and Empirical Targets for Early Period (1980)

Parameter	Value	Target	Data	Model
ϕ_g	0.2016	HS wage bill share (goods)	0.1822	0.1822
ϕ_o	0.4933	HS wage bill share (other)	0.4592	0.4592
ϕ_s	0.1977	HS wage bill share (services)	0.1772	0.1772
ζ_g	0.1307	Male wage bill share (goods)	0.8314	0.8314
ζ_o	0.3350	Male wage bill share (other)	0.6057	0.6057
ζ_s	0.2881	Male wage bill share (services)	0.6468	0.6468
β_G	0.2190	Mkt hours share (goods)	0.4349	0.4349
β_O	0.1988	Mkt hours share (other)	0.3985	0.3985
ω	0.2373	HP share single M-LS	0.2359	0.2359
b_l^w	1.3172	HP share single F-LS	0.4269	0.4269
b_h^w	1.6948	HP share single F-HS	0.3855	0.3855
b_h^m	1.4272	HP share single M-HS	0.2652	0.2652
$\tilde{u}_{hh}$	0.0653	Single rate F-HS	0.2618	0.2731
$\tilde{u}_{hl}$	0.0627	Single rate F-LS	0.1312	0.1451
		Single rate M-HS	0.2137	0.1953
		Single rate M-LS	0.1784	0.1599
$\tilde{u}_{lh}$	0.1046	Couple share HH	0.1234	0.1109
$\tilde{u}_{ll}$	0.1044	Couple share LL	0.7003	0.7301
		Couple share LH	0.1249	0.1131
γ	16.4782	Shenhav semi-elasticity	-0.3000	-0.3000

share of women aged 22-44 who have never been married. We perturb the gender wage ratio for each skill group within the model, holding prices fixed in partial equilibrium, and ask the model to generate an equivalent decline in the marriage share.

As can be seen in Table 3, the model fits the data well, even for the second set of target moments, for which the calibration is over-identified. Furthermore, the parameter values are in line with expectations. For example, women have higher home productivity than men within both skill groups. Additionally, the non-substitutable service sector O is the most skill-intensive, while the goods sector G is the most male-intensive.

5.3 Calibration of Model to the Later Period

We re-calibrate the model to 2010, allowing only the following parameters to change: the college shares for men and women (h_M and h_F), which we observe directly in the Census,

as well as the sector productivity terms (A 's) and the couple-type specific preference terms ($\tilde{u}$'s). The latter are calibrated to jointly match an updated set of targets from the 2010 ACS. Table 4 shows the parameters and model fit. Consistent with theories of structural transformation, we find growth in the relative productivity of goods vs. services. Within services, the growth of productivity in S relative to O is consistent with marketization.

Table 4: Calibrated Parameters and Empirical Targets for Late Period (2010)

Parameter	Value	Target	Data	Model
A_g	1.2795	Mkt hours share (goods)	0.3194	0.3194
A_o	0.7234	Mkt hours share (other)	0.4889	0.4889
A_s	1.0271	HP share of total time	0.4238	0.4238
$\tilde{u}_{hh}$	0.0436	Single rate F-HS	0.2738	0.2840
$\tilde{u}_{hl}$	0.0603	Single rate F-LS	0.2567	0.2841
		Single rate M-HS	0.2705	0.2474
		Single rate M-LS	0.3222	0.2987
$\tilde{u}_{lh}$	0.1432	Couple share HH	0.2382	0.2061
$\tilde{u}_{ll}$	0.0530	Couple share LL	0.5347	0.5769
		Couple share LH	0.0983	0.0943

6 Quantitative Exercise

Using our calibrated model, we quantify the role of structural transformation in shaping marriage patterns by skill. First, we show that structural transformation affects marriage incentives differently across skill groups: economic forces increased the marital surplus among high-skill couples between 1980 and 2010, and lowered it among low-skill couples. Second, through counterfactual analysis, we show that structural transformation contributed to the widening marriage gap between college and non-college individuals, as depicted in Figure 1.

6.1 The Role of the Marital Surplus

Guided by Equation (23), we decompose changes in the log-odds ratio of marriage by potential couple type into both economic forces which we model, as well as preferences which we take

to be exogenous. As can be seen in Table 5, the model-implied change in the log-odds ratio of marriage evolves differently across potential couple types, but declines for the most empirically prevalent couple pairings: those with the same education levels, HH and LL . However, the contribution of economic forces (structural transformation and the increase in the supply of educated workers) affect marriage incentives differently across these couple types, as shown by the column “Surplus Channel.” The marital surplus increases for HH couples, as well as couples in which the wife is more educated than the husband (HL). However, the economic forces negatively impact marriage probabilities among LL couples, as well as couples where the wife is less educated than the husband (LH). Via the surplus channel, structural transformation explains approximately one quarter of the decline in the log-odds ratio of marriage among LL couples.

Table 5: Log-Odds Decomposition of Marriage Rate Changes

Couple	Δ Log-Odds	Surplus Channel	Preference Channel	% Surplus
HH	-0.182	+0.223	-0.405	—
HL	+0.201	+0.240	-0.039	119.2
LH	+0.025	-0.288	+0.313	—
LL	-0.924	-0.247	-0.677	26.7

Notes: Decomposition follows Equation 23. Δ Log-odds computed from model π . Surplus channel = $\gamma \times \Delta \log(C^{mar}/(C_f^a + C_m^a))$. Preference channel = $\Delta \log \tilde{u}$. Couple types order female skill first and male skill second.

6.2 Decomposing the Aggregate Decline in Marriage Rates

The results in Table 5 show that structural transformation had different effects on the incentive to marry across education groups. Next, we examine to what extent these induced changes in the marital surplus can account for the observed decline in marriage and the widening education gap.

Table 6 shows the aggregate share of individuals who are married by skill group. The first two columns show the rates for the model calibrated to 1980 and 2010, respectively. The latter two columns show the rates under two counterfactual scenarios. Both start from the

1980 calibration, but one allows only the sector productivity terms and education shares to evolve to 2010 values, while the other allows only the marital preference parameters to evolve to the 2010 values. We call the former (A 's + h) our structural transformation counterfactual.

Table 6: Marriage Shares by Education: Counterfactual Analysis

Education	Early	Late	A 's + h	$\tilde{u}$'s
College+	0.771	0.733	0.779	0.730
No BA	0.848	0.708	0.809	0.758
Difference	-0.077	+0.025	-0.030	-0.028

Notes: The “Early” and “Late” columns show the share of individuals by skill group who are married in the model calibrated to 1980 and to 2010, respectively. The “ A 's + h ” column shows the marriage rates under a structural transformation counterfactual where only the productivity parameters and education shares change from 1980 levels. The “ $\tilde{u}$'s” columns shows a counterfactual where only the preference terms change from 1980 levels. The bottom row shows the difference in marriage rates between College+ and No BA.

From the early to late period, both education groups see a decline in marriage rates, but the decline is much bigger for those without college. The gap in marriage rates between college and non-college individuals starts at -7.7 p.p. but flips signs over this period, reaching +2.5 p.p. Overall, the College+ minus No BA marriage gap grows by 10.2 p.p. between 1980 and 2010. The structural transformation counterfactual alone can account for 4.7 p.p. (or 46%) of this widening education gap in marriage rates. Interestingly, structural transformation leads to, if anything, a small increase in the marriage share for college-educated individuals, consistent with the increase in the marital surplus reported in Table 5. By contrast, the structural transformation forces alone generate a 3.9 p.p. decline in the marriage share for the less-educated, out of a total observed 14.0 p.p. decline.

7 Conclusion & Next Steps

This paper has analyzed the connection between two profound changes in American society: the shift from an industrial economy to a service economy, and the decline of marriage. Structural transformation narrows the gender wage gap, especially for non-college workers, and provides market substitutes for services that were traditionally produced within the

home. We develop a Beckerian model of household formation in which both of these effects contribute to the decline of marriage, by reducing men’s comparative advantage for market production and women’s comparative advantage for home production. After calibrating the model to match the evolution of time use, employment, and marriage among American households, we find that structural transformation forces explain approximately one-third of the 14 percentage point decline in marriage among Americans without college, and close to one-half of the divergence between college and non-college marriage rates.

We view this draft as presenting preliminary results for the impact of structural transformation on marriage. In future versions, we intend to endogenize households’ choice of skill level. We also plan to make greater use of data on households’ consumption behavior, to provide more evidence on the rising use of market substitutes for home production. Further exploration of the model will allow us to assess the general equilibrium feedback between the decline of marriage and the rise of services: because single households consume more home production substitutes, the rising single share of households pushes even greater marketization of home services. Finally, we believe that the same methodology we develop in this paper could be used to analyze global trends. De-industrialization and collapsing marriage and fertility rates have taken hold in many countries around the world. Our paper provides an approach to measure how tightly connected these trends are.

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A Theoretical Appendix

Deriving the consumption gains from marriage Equation (21) follows from writing $\log\left(\frac{C_i}{C_{F,i}^a + C_{M,i}^a}\right) = -\log\left(\frac{C_{F,i}}{C_i} + \frac{C_{M,i}}{C_i}\right)$ and substituting in expressions for consumption in terms of resources and prices. For Equation (22), consider $\frac{P_i}{P_{F,i}}$; the process to derive $\frac{P_i}{P_{M,i}}$ is identical. First note that

$$\left(\frac{P_i}{P_{F,i}}\right)^{1-\sigma} = \frac{\beta_G P_G^{1-\sigma} + \beta_O P_O^{1-\sigma} + \beta_{HS} P_{HS,i}^{1-\sigma}}{\beta_G P_G^{1-\sigma} + \beta_O P_O^{1-\sigma} + \beta_{HS} P_{HS,F,i}^{1-\sigma}} = 1 + \beta_{HS} \frac{P_{HS,i}^{1-\sigma} - P_{HS,F,i}^{1-\sigma}}{P_{F,i}^{1-\sigma}}. \quad (24)$$

Substituting in $\vartheta_{HS,F,i} = \beta_{HS} \frac{P_{HS,F,i}^{1-\sigma}}{P_{F,i}^{1-\sigma}}$ yields

$$\left(\frac{P_i}{P_{F,i}}\right)^{1-\sigma} = 1 + \vartheta_{HS,F,i} \left[\left(\frac{P_{HS,i}}{P_{HS,F,i}}\right)^{1-\sigma} - 1 \right]. \quad (25)$$

Next consider $\frac{P_{HS,i}}{P_{HS,F,i}}$. Again we use the fact that the home service price index differs only in the home production part $P_{HP,i}$, not the market substitutes part P_S :

$$\begin{aligned} \left(\frac{P_{HS,i}}{P_{HS,F,i}}\right)^{1-\xi} &= \frac{\omega P_S^{1-\xi} + (1-\omega) P_{HP,i}^{1-\xi}}{\omega P_S^{1-\xi} + (1-\omega) P_{HP,F,i}^{1-\xi}} \\ &= 1 + (1-\omega) \left[\left(\frac{P_{HP,i}}{P_{HS,i}}\right)^{1-\xi} \left(\frac{P_{HS,i}}{P_{HS,F,i}}\right)^{1-\xi} - \left(\frac{P_{HP,F,i}}{P_{HS,F,i}}\right)^{1-\xi} \right]. \end{aligned} \quad (26)$$

Now we do some algebra and substitute in $\mu_{F,i} = \omega \frac{P_S^{1-\xi}}{\omega P_S^{1-\xi} + (1-\omega) P_{HP,F,i}^{1-\xi}} = 1 - (1-\omega) \left(\frac{P_{HP,F,i}}{P_{HS,F,i}}\right)^{1-\xi}$:

$$\left(\frac{P_{HS,i}}{P_{HS,F,i}}\right)^{1-\xi} = \frac{1 - (1-\omega) \left(\frac{P_{HP,F,i}}{P_{HS,F,i}}\right)^{1-\xi}}{1 - (1-\omega) \left(\frac{P_{HP,i}}{P_{HS,i}}\right)^{1-\xi}} = \frac{\mu_{F,i}}{\mu_i}. \quad (27)$$

We therefore obtain $\frac{P_{HS,i}}{P_{HS,F,i}} = \left(\frac{\mu_{F,i}}{\mu_i}\right)^{\frac{1}{1-\xi}}$. Substituting this back into Equation (25) gives us Equation (22).